

Product name	Product discription	Product link	product unit price (AUD)	Product amount(AUD)	Product application in project	shipping cost
ESP32 development board	a powerful, low-cost, and highly integrated microcontroller widely used for Internet of Things (IoT) projects. It features built-in Wi-Fi and Bluetooth capabilities, a dual-core processor, and multiple GPIO pins for connecting sensors, motors, and displays.	esp	10.25		This micro controller connects to the servos touchboard and OLED screen. with the output pins	3.02
polymer Lithium Ion Battery (LiPo) 3.7V 1500mAh	a thin, lightweight, and rechargeable power source commonly used in low-drain, portable electronics and microcontroller projects (like Adafruit Feathers). It provides a nominal 3.7V output and holds a capacity of 1500mAh, about 5.5 WHS	lithium bat	8.02		This battery powers all the electronics components in the robot it is hooked to the tp4056 battery charges and has a good compacity	3.4
TP4056 Charging Module with Battery Protection	a compact, dual-function electronic board designed for safely managing single-cell 3.7V lithium-ion or lithium-polymer batteries	charging	0.21		the charging module to charge the lithium battery	3.51
Capacitive Touch Sensor	Capacitive touch sensors are advanced input devices that register touch by detecting changes in electrical capacitance caused by a conductive object, such as the human body	touch sensor	1.95		This sensor will the input part into the robot, and can trigger some fuctions output by the OLED and servos	1.714
9g micro servo	an extremely compact, lightweight, and affordable motor used in small-scale robotics, RC planes, and DIY electronics. Weighing only 9 grams, it typically provides up to 1.6 of torque and offers 180-degree rotation	servo	5.36		6 degrees of freedom given by these servos and the 5 legs 2 servos on each leg	1.714
jumperwires	10cm male-to-male jumper wires. Pre-terminated with an assortment of colors, these 30 AWG wires are perfect for prototyping and connecting between breadboards and female headers.	jumperwires	NA	Own at home	the jumper wires connect the diffrent electronics components	0
OLED	a compact, monochrome display widely used in DIY electronics and microcontrollers (like Arduino and Rasperry Pi Pico). It features a (128 \times 64) pixel resolution, delivers excellent contrast with true blacks, and uses I2C or SPI interfaces for easy integration	OLED	6.7		the OLED provides a output for expression and information, for exp tempreture , weather and googly eyes	1.714
AAA battery	a small, cylindrical, single-cell dry battery	AAA bat		3 need 4 but sells in budles of 8	used to power servos sit in battery box	0
Toggle Switch 3Pins Straight Feet Handle	High 5mm Spacing: Ensures stable and secure PCB mounting for reliable performance.3-Pin Configuration: Simplifies wiring and enhances compatibility with common circuit layouts.3A 250V Rating: Provides safe and efficient power control for various electronic devices. SS12D10/SS12D06 Model: Standardized toggle switches suitable for a wide range of applications.Straight Feet Handle Design: Offers easy installation and improved accessibiliity on circuit boards.	switch	1.9	1 needed but sells in bundels of 10	This is so the battery isnt running all the time and also acts as a saftey feature	3.72
DHT 22	The DHT22 is a basic, low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and spits out a digital signal on the data pin (no analog input pins needed).	DHT	4.61		To measure the tempreture and humidity of where it is, do this so that it can start to give warning or tips to open the windows or turn of airconditioner	1.37
10k resistor	a standard, general-purpose component used in digital electronics to ensure an input pin maintains a stable voltage (HIGH) when no other signal is active	10k	2.75	4 needed but sold in packs of 20	use in robot to stabilise voltage and signal on DHT 22 and OLED	1.714
battery box 4*AA	an enclosed plastic container designed to hold four AAA batteries in a series configuration	AAA	4.25		Powers 10 servos	1.714
SUBTOTAL			49			23.59
Total: 72.59						